

Course Information

Course: Course Name

Course Code: 12345

Credit Hours: 3

Textbook: Louden, K. C., & Lambert, K. A. (2011). *Programming languages: principles and practices*. Cengage Learning.

Lecturer Information

Name: Dr. Homer Simpson

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Course description

Principles and techniques of high level programming language specification, including syntax and semantics, and implementation methods. Languages are considered as formal constructs representing computable functions and the course presents basic results of computability theory. Methods of recognition for regular and context free languages, the functional alternative to the state based imperative paradigm, use of parse trees as control constructs, and concurrent programming synchronization primitives are also covered.

Topics

- Generate data, create data frame (structure)
 - Data science knowledge and skills (background)
 - Data science careers (side focus)
 - Describe Data Science and the skill sets needed to be a data scientist.
 - Data acquisition (sql, csv, reading, loading, Web Scraping, etc.) and wrangling
 - Exploratory data analysis (visualization and description)
 - Communication (Rmd, shiny, and slidy)
 - Version control (git)
 - Tidyverse
 - BigData
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Learning Outcomes

By the end of this course, participants will be able to:

- Discuss database management system concepts

- Design a database: This will include design philosophies such as E-R and Relational Algebra, along with studies of normal forms for databases
 - Write SQL queries
 - Explain Physical Representation. This will cover the basics of how the information in a database is stored and accessed on various systems
 - Explain advanced NoSQL and Big data frameworks
 - Apply the database concepts to a real-world project
 - Demonstrate knowledge of data science ethics in forum discussions.
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Course Evaluation and Grading Policies

Grades are based on: Attendance/Participation (10%), exams (2 @ 10%), homework (20%), projects (40%), and final exam (10%).

The grade of "I" (incomplete) can be given ONLY when a student, who is doing otherwise acceptable work (passing grade) and cannot complete a part of work (e.g., the final exam) because of documented illness or other conditions beyond the student's control. In the latter case, the student must discuss the situation with the instructor and complete an application form from the department before the part of the work is due or as soon as the circumstances are known.

Essential Equipment

Typical technical suggestions for users are:

- Computer or Laptop: A reliable computer or laptop is the foundation of your setup. Choose one that meets your requirements in terms of processing power, storage capacity, and portability.
- Operating System: Install a suitable operating system such as Windows, macOS, or Linux, depending on your preference and software compatibility.
- RAM (Random Access Memory): Opt for at least 8GB of RAM to ensure smooth multitasking and efficient performance. However, if you plan to run resource-intensive applications or perform data-intensive tasks like video editing or 3D rendering, consider getting 16GB or more.
- Storage Drive: Have a sufficient storage drive to store your files and applications. A solid-state drive (SSD) is recommended for faster boot times and improved overall system performance. Aim for a minimum of 256GB, but consider larger capacities if you work with large files or require extensive storage.
- Monitor: Invest in a good-quality monitor with a suitable size and resolution for your needs. Ensure it has the necessary ports to connect to your computer.
- Keyboard and Mouse: Choose a comfortable keyboard and mouse for easy navigation and typing. Wired or wireless options are available, depending on your preference.
- Internet Connectivity: Reliable internet connectivity is essential for accessing online resources, cloud storage, and software updates. Ensure you have a stable internet connection either through Ethernet or Wi-Fi.
- Speakers or Headphones: Consider having speakers or headphones for audio output, depending on your preference. This will enable you to listen to music, watch videos, or participate in online meetings.

- Webcam: If you plan to engage in video conferencing or online meetings, a webcam is necessary for video communication.
 - Surge Protector or Uninterruptible Power Supply (UPS): Protect your computer and equipment from power fluctuations or outages by using a surge protector or UPS.
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